MALLA REDDY UNIVERSITY

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

SPIDER ROBOT FOR HAZARDOUS AREA INSPECTION

An Application Development Project Report

B.Tech II Year
ECE-ALPHA

Academic Year
2026 – 2027

1. ABSTRACT

Inspecting collapsed structures, industrial areas, disaster zones and fire-prone spaces puts human lives at risk. This project presents a **four-legged spider robot** for remote inspection and hazard detection, reducing the need for people to enter dangerous environments.

An **ESP32-CAM** serves as the controller and captures real-time images, an **ultrasonic sensor** detects obstacles, and **servo motors** drive the legs for mobility over uneven terrain. A **buck converter** supplies regulated power to all components.

Goal: Design & develop a low-cost, compact, remotely deployable robot for safe hazardous-area inspection.

2. PROBLEM & OBJECTIVES

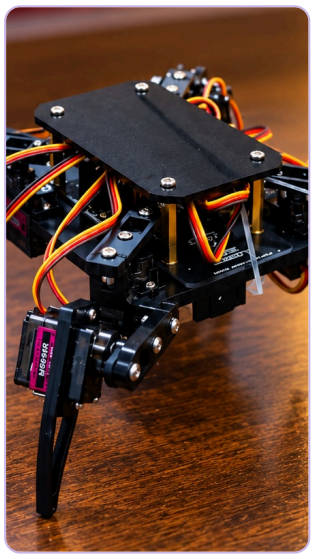
Problem Statement

Human entry into hazardous and inaccessible areas is unsafe, and such spaces are hard to reach and assess remotely.

Objectives

- Build a servo-driven four-legged robot for uneven terrain
- Detect and avoid obstacles using an ultrasonic sensor
- Capture surroundings with ESP32-CAM for remote analysis
- Provide regulated power using a buck converter

3. OVERVIEW & KEY FEATURES

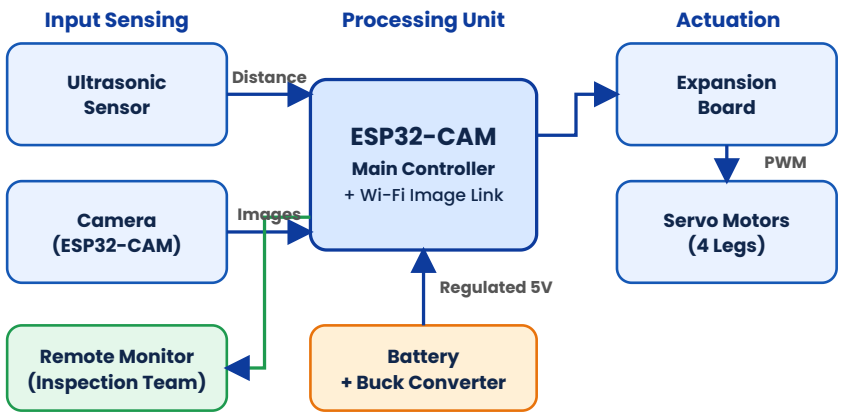


Developed Spider Robot Prototype

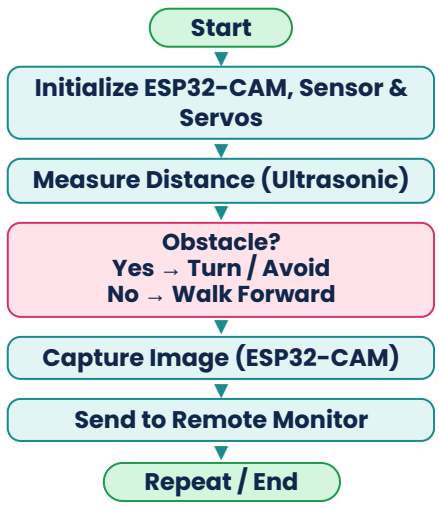
Key Features

- Four-Legged Servo Locomotion
- Ultrasonic Obstacle Detection
- Real-Time Image Capture (ESP32-CAM)
- Remote Monitoring Support
- Regulated Power (Buck Converter)

4. SYSTEM ARCHITECTURE



5. FLOWCHART



6. HARDWARE & SOFTWARE

Hardware

- ESP32-CAM Module
- Ultrasonic Sensor
- Servo Motors (Legs)
- Buck Converter
- Expansion Board
- Battery Pack
- Chassis & Legs

Software

- Embedded C/C++ Firmware
- ESP32 Camera Library
- Servo Control Library
- Wi-Fi Web Server
- Browser / Mobile Viewer

7. WORKING PRINCIPLE

Sensing — Ultrasonic checks distance to obstacles

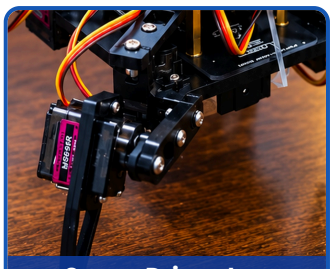
Processing — ESP32-CAM decides path & gait

Walking — Servos move the four legs

Imaging — Camera captures surroundings

Inspection — Remote team analyses images

8. PROTOTYPE & MONITORING



Servo-Driven Leg Mechanism

Robot enters the hazardous zone

ESP32-CAM captures images

Team inspects remotely & stays safe

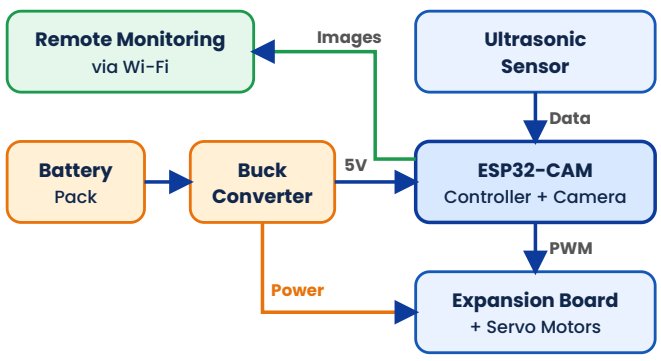
9. APPLICATIONS

- Collapsed-structure search
- Industrial hazard inspection
- Fire-prone zone survey
- Confined-space inspection
- Educational robotics

10. EXPECTED RESULTS

- Stable four-legged walking over uneven terrain
- Reliable obstacle detection and avoidance
- Clear image capture of surroundings
- Remote inspection without human exposure to risk
- Low-cost, compact and portable design

11. BLOCK DIAGRAM



12. CONCLUSION & FUTURE SCOPE

Conclusion

A low-cost, compact spider robot for remote hazardous-area inspection, combining legged mobility, obstacle sensing and image capture.

Future Scope

Live video streaming, environmental/gas sensors, autonomous navigation and advanced image processing for hazard identification.

Roll Numbers & Student Names

K. GOPICHAND — 2511EC010040
P. NIRGUNA REDDY — 2511EC010051
T. YUVARAJSINGH — 2511EC010056

Guide Name:
Ms. S. Mrudhula
(Assistant Professor)